

Preeclampsia risk prediction from prenatal cell-free DNA screening

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Preeclampsia is characterized by placental dysfunction and results in significant morbidity, but reliable early prediction remains challenging. We investigated whether clinically obtained prenatal cell-free DNA (cfDNA) screening (PDNAS) using whole-genome sequencing (WGS) data can be leveraged to predict preeclampsia risk early in pregnancy (≤ 16 weeks). Using 1,854 routinely collected clinical PDNAS samples (median, 12.1 weeks) with low-coverage ($0.5\times$) WGS data, we developed a framework to quantify maternal and fetal tissue signatures using nucleosome accessibility, revealing early placental and endothelial dysfunction. These signatures informed a prediction model for preeclampsia risk, which achieved a validation performance of 0.85 area under the receiver operating characteristic curve (AUC) (81% sensitivity at 80% specificity) for preterm phenotypes several months prior to disease onset in a separate cohort of 831 consecutively collected samples, and subsequently confirmed in an external cohort of 141 samples (AUC 0.84, 79% sensitivity). We demonstrate that assessment of cfDNA nucleosome accessibility from early-pregnancy cfDNA sequence data enables the detection of early placental and endothelial-tissue aberrations and may aid in the determination of preeclampsia risk.

Preeclampsia, a hypertensive disorder affecting 2–8% of pregnancies, contributes substantial morbidity and mortality to the maternal–fetal dyad, particularly when delivered preterm^{1–4}. Prediction of preeclampsia before disease onset has remained challenging, with many approaches relying on binary risk classification on the basis of clinical factors alone or in combination with circulating biomarkers^{5,6}. The performance of these methods, however, is suboptimal in many cases^{7,8}. The placenta has a well-defined role in the developmental origins of preeclampsia^{9–11}.

Given that the placenta is inaccessible during pregnancy, non-invasive approaches are essential to identify molecular features indicative of preeclampsia risk, improve the understanding of the underlying biology, and develop strategies to mitigate the risks of preeclampsia.

Constitutive shedding of placenta-derived material, such as cell-free RNA (cfRNA) and cfDNA, begins early in pregnancy^{12–14}. Recent studies investigated preeclampsia prediction during early pregnancy using circulating cfRNA, long-molecule cfDNA, and cfDNA methylation^{15–18};

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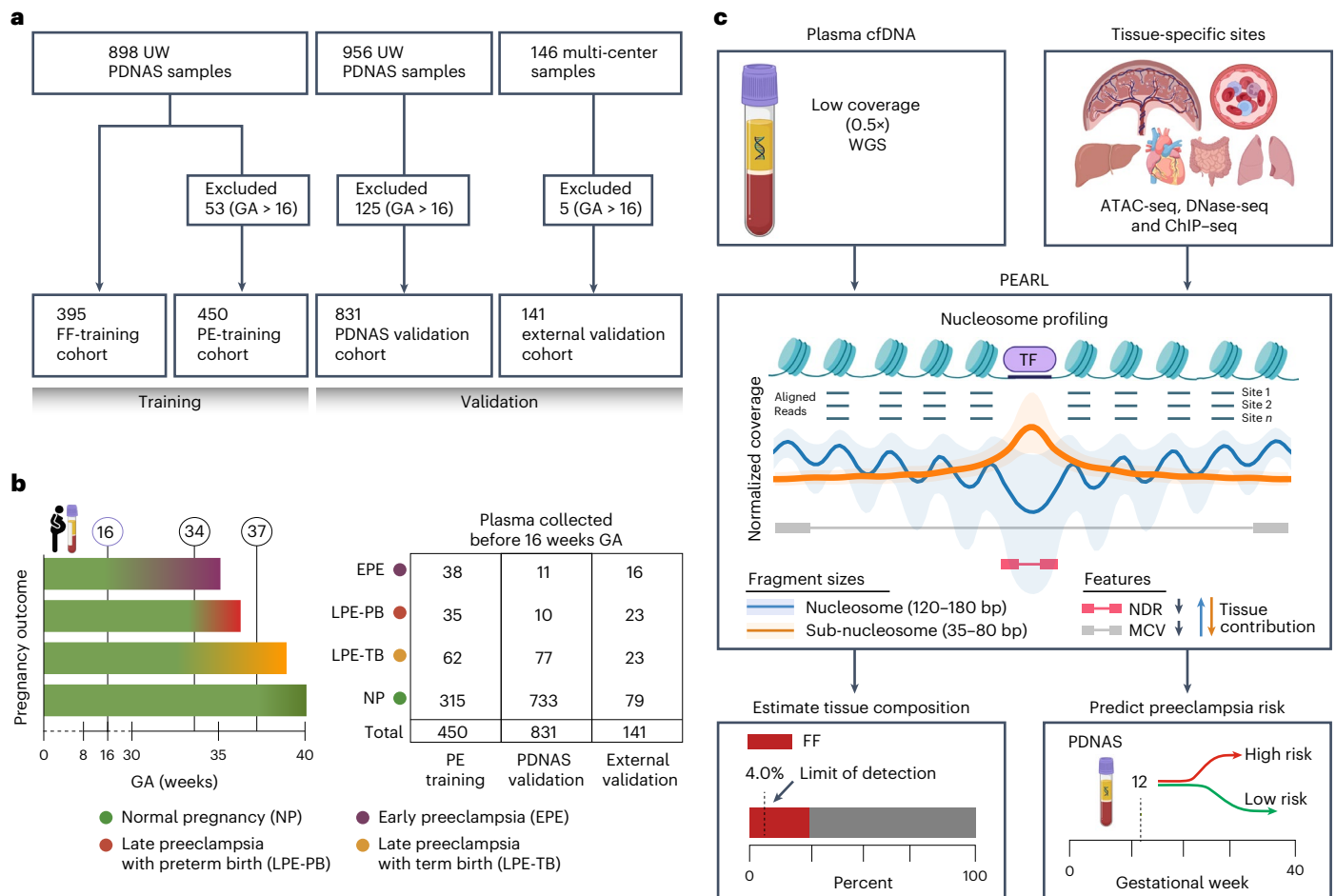


Fig. 1 | Study design to predict maternal–fetal tissue composition and preeclampsia risk using cfDNA sequencing data. a, Flow chart of clinical PDNAS samples consisting of separate and distinct training and validation cohorts. A total of 1,854 PDNAS samples from the UW clinical laboratory were included. The training cohorts were the FF-training cohort, consisting of 375 PDNAS samples with XY fetuses and 20 from non-pregnant healthy females, and the PE-training cohort, including a random sampling of PDNAS samples collected at ≤ 16 weeks gestation from 1 January 2019 to 31 December 2020 with enrichment for additional preeclampsia (PE) cases from 1 January 2021 to 25 April 2023 (total $n = 450$). The validation cohorts were the PDNAS validation cohort ($n = 831$) and the multicenter external cohort ($n = 141$). The PDNAS validation cohort consists of clinical PDNAS samples collected consecutively at ≤ 16 weeks gestation from 1 May 2017 to 31 December 2018 that met the inclusion criteria ($n = 831$). The multicenter external cohort comprises samples collected at ≤ 16 weeks gestation from two biobanks that were sequenced at an independent clinical laboratory ($n = 141$). GA, gestational age (weeks). **b**, Schematic of preeclampsia outcome classification. The x-axis indicates GA (weeks), and the y-axis indicates outcome. Colors show the time from clinical onset of hypertension to delivery. The box (right) indicates

the number of samples included in the training and validation cohorts. The NP group in the PDNAS validation cohort includes 44 samples from preterm births that were not complicated by PE. **c**, PEARL framework for estimating the tissue contribution in plasma cfDNA from PDNAS data using tissue-specific chromatin profiling from low-coverage WGS ($\sim 0.5\times$) to estimate FF and predict PE risk. Chromatin profiles are generated at tissue-specific accessible sites obtained from publicly available accessibility datasets. Chromatin features include nucleosome and sub-nucleosome profiles generated using fragments sizes restricted to 120–180 bp and 35–80 bp, respectively. To overcome low coverage, sequence reads were aggregated across 10,000 tissue-specific accessible sites. On the basis of the distance from accessible sites, features such as MCV ($\pm 1,000$ bp) and NDR (± 30 or 75 bp) were extracted from the profiles. These features reflect tissue contributions, which were then used to train supervised models to estimate the FF and predict PE risk. **b** and **c** were created using BioRender (<https://BioRender.com/i89p411>). ATAC-seq, Assay for Transposase-Accessible Chromatin using sequencing; DNase-seq, DNase I hypersensitive sites sequencing; ChIP-seq, chromatin immunoprecipitation and sequencing.

however, these methods require specialized assays, increasing costs and complicating clinical implementation. By contrast, sequencing of first-trimester maternal plasma cfDNA, already performed for prenatal screening of aneuploidy, is increasingly part of routine prenatal care¹⁹, albeit mostly in high-income countries²⁰. A tool for early preeclampsia prediction that seamlessly integrates into routine practice could substantially reduce the wide-ranging adverse sequelae of preeclampsia.

cfDNA fragments are preferentially protected from degradation by nucleosomes, reflecting the chromatin organization of their tissue of origin^{21–26}. Although clinical prenatal cfDNA screening (PDNAS) assays routinely determine the placental-derived fraction, or fetal fraction (FF)¹⁹, the assessment of tissue-specific nucleosome footprints in

cfDNA for the early prediction of pregnancy-specific disorders, such as preeclampsia²⁷, has not been explored. Using sequence data generated through PDNAS in singleton gestations, we developed a new computational framework called preeclampsia early assessment of risk from liquid biopsy (PEARL) to accurately quantify the maternal–fetal tissue composition in plasma cfDNA and to classify the risk of severe phenotypes of preeclampsia months before clinical onset.

Results

Study design using clinical PDNAS data

Using low-coverage WGS data from 1,854 clinical plasma-cfDNA samples for PDNAS (mean, $0.52\times$ coverage; interquartile range (IQR),

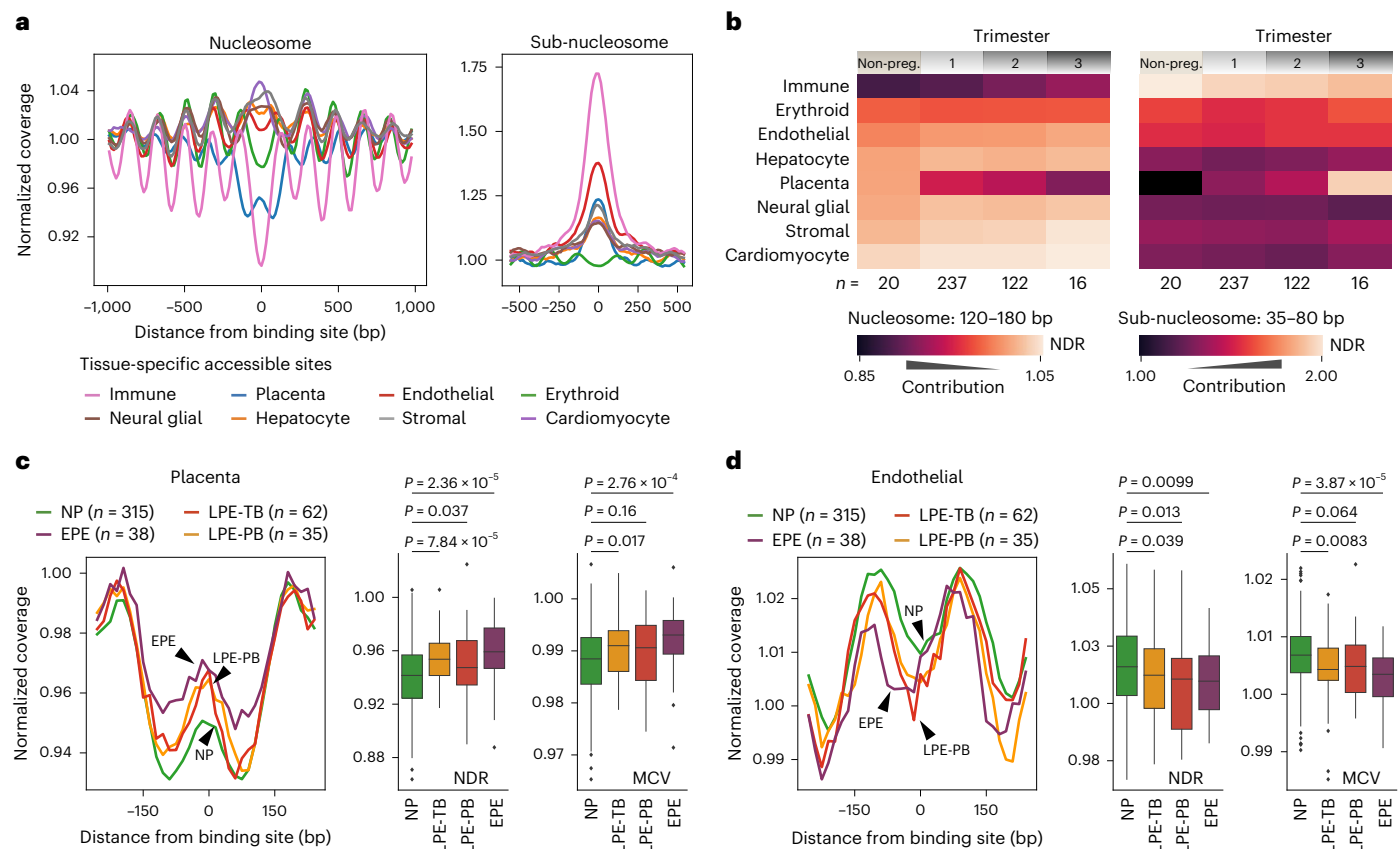


Fig. 2 | Nucleosome patterns in cfDNA discern maternal–fetal tissue composition and early tissue dysfunction in those who develop preeclampsia.

a, Nucleosome profile (left) and sub-nucleosome profile (right) of tissue-specific accessible sites derived from single-cell ATAC-seq (scATAC-seq) in 395 PDNAS samples (FF-training cohort). Immune (pink) and placenta (blue) have lower and higher NDR values in nucleosome and sub-nucleosome profiles, respectively, indicating higher tissue contribution. **b**, Heatmap of normalized coverage values at NDRs for tissue-specific nucleosome (left) and sub-nucleosome (right) features derived from scATAC-seq for 20 non-pregnancy and 375 pregnancy PDNAS samples, stratified by trimesters (FF-training cohort). **c**, Placental

nucleosome profile (left) and NDR and MCV (right) for samples from the PE-training cohort ($n = 450$). Higher NDR and MCV values in samples with a PE outcome indicate decreased placental contribution. **d**, Endothelial nucleosome profile (left) and NDR and MCV (right) for samples from the PE-training cohort ($n = 450$). Lower NDR and MCV values in samples with a PE outcome indicate increased endothelial contribution. Boxes represent the IQR (25th–75th percentiles), with the horizontal line indicating the median. The whiskers extend to $1.5 \times$ IQR on either side, with dots representing outliers. Statistical analysis was done using the two-sided Mann–Whitney U test.

0.36–0.64) from the University of Washington (UW), we developed a framework to accurately quantify tissue contributions to maternal plasma and predict preeclampsia risk months before clinical onset (Fig. 1a and Extended Data Table 1). Preeclampsia was categorized into clinically relevant subtypes on the basis of gestational age at onset and delivery: early preeclampsia (EPE) (onset, ≤ 34 weeks, and delivery, < 37 weeks); late preeclampsia (LPE) with preterm birth (LPE-PB) (onset, > 34 weeks, and delivery, < 37 weeks); and LPE with term birth (LPE-TB) (onset, > 34 weeks, and delivery, ≥ 37 weeks) (Fig. 1b, Extended Data Fig. 1, Extended Data Table 1, Supplementary Table 1, and Methods). We applied the PEARL framework, which uses the Griffin tool²², to assess tissue-specific nucleosome profiles in cfDNA and quantify features of the nucleosome-depleted region (NDR, ± 30 and ± 75 bp) and the mean coverage around the site (MCV, $\pm 1,000$ bp) (Fig. 1c and Methods). These features were used in machine-learning models of the PEARL framework for (1) estimating the maternal–fetal tissue composition and FF and (2) predicting the risk of preeclampsia.

Early tissue dysfunction in those who develop preeclampsia

To study early tissue dysfunction in individuals who eventually developed preeclampsia, we used PEARL to analyze PDNAS samples, demonstrating its ability to discern cfDNA derived from distinct tissues,

including placenta²⁸ (Fig. 2a, Extended Data Fig. 2, and Methods). We confirmed that the contribution of placental tissue in PDNAS samples was detected in samples from pregnant people and increased in each trimester (Pearson's $r = -0.42$, $P = 2.28 \times 10^{-18}$), whereas the contribution of immune and endothelial cells decreased by trimester ($P < 0.020$; Fig. 2b and Extended Data Figs. 2–6). Next, we investigated whether early-pregnancy (≤ 16 weeks) PDNAS data could reveal tissue-contribution differences that are unique to individuals who developed preeclampsia. We used PDNAS samples from 450 individuals (PE-training cohort), collected at a median of 12.2 weeks (range, 8.1–16.0 weeks); 135 of these individuals eventually developed preeclampsia, and 315 had normal pregnancies (Fig. 1a,b, Extended Data Table 1, and Methods). We observed that the placental-tissue contribution was significantly lower in those who developed preeclampsia than in those with normal pregnancies (Mann–Whitney U , $P < 0.037$; analysis of covariance (ANCOVA), $P \leq 0.03$; Fig. 2c, Extended Data Fig. 7, and Methods). By contrast, we found that the contribution of endothelial tissue was higher in those with preeclampsia than in those with normal pregnancies (Mann–Whitney U , $P \leq 0.039$; ANCOVA, $P \leq 0.012$; Fig. 2d and Extended Data Fig. 7). These results identify the presence of early placental dysfunction in individuals who develop preeclampsia, and may also reflect underlying maternal endothelial dysfunction^{29–31}.

Sex-independent FF varies by preeclampsia severity

Differences in maternal–fetal tissue composition in plasma cfDNA are not well understood in the context of preeclampsia. A critical early step in PDNAS is accurate quantification of the FF, which can be influenced by fetal sex, genotype, and fragment length^{32–34}. We sought to overcome these potential biases by relying on placental-specific signatures to determine the FF. First, we used the PEARL framework to develop a new model to estimate the FF, independent of sex, genotype, or differences in fragment length^{32,35,36}, by integrating 1,239 features across placental and maternal tissue types, as well as 377 transcription factors (Methods). The model was developed using 395 PDNAS samples from 375 pregnant women with male (XY) fetuses, whose FF had been determined by Y-chromosome analysis, and 20 non-pregnant women (FF-training cohort; Fig. 1a and Methods). The training performance showed a strong positive correlation (Pearson's $r = 0.90$), with a limit of detection of 0.039 (range, 0.039–0.042) (Fig. 3a, and Methods). We validated the model's performance on a publicly available dataset ($n = 70$) in which the FF was determined by maternal–fetal genotyping for fetuses of both sexes³⁷ (Pearson's $r = 0.97$; Fig. 3b, Extended Data Fig. 8, and Supplementary Table 2), and on 238 XY fetuses in the PE-training cohort (Pearson's $r = 0.79$; Fig. 3c).

We next assembled a cohort of 956 routine PDNAS samples, collected consecutively from May 2017 to December 2018 (PDNAS validation cohort), approximating an all-comers population including individuals with pre-existing hypertension, diabetes, preterm birth, and other maternal medical conditions (Fig. 1a and Methods). First, we confirmed the predicted FF in samples from XY fetuses in this cohort ($n = 462$, Pearson's $r = 0.85$) (Fig. 3d) by applying the final model previously trained on the FF-training cohort. We also observed that, in all three cohorts, the estimated FF was significantly lower for individuals who developed EPE than for those with normal pregnancies (Mann–Whitney U , $P \leq 0.044$; Fig. 3e and Extended Data Fig. 7). Furthermore, we observed that the estimated FF significantly increased across gestation in normal pregnancies; however, it remained lower throughout pregnancy in those with EPE (Mann–Whitney U ; $P = 0.0088$ versus 0.91; Fig. 3f), indicative of persistent detectable tissue dysfunction throughout pregnancy.

PEARL predicts preeclampsia risk using PDNAS data

We next investigated whether cfDNA nucleosome features can expand the utility of early-pregnancy PDNAS (≤ 16 weeks) to predict preeclampsia risk. Using the PEARL framework, we used 450 PDNAS samples (PE-training cohort) to train a machine-learning binary classifier to predict preeclampsia risk (Fig. 1a,b). The model incorporates two feature sets: (1) placental- and endothelial-tissue-specific nucleosome and subnucleosome features (NDR and MCV) and the PEARL-estimated FF; and (2) blood pressure and body mass index (BMI), both of which are easily ascertained around the time of PDNAS (Methods). The model achieved an overall training performance of 0.76 AUC (confidence

interval (CI), 0.70–0.82) for predicting preeclampsia requiring preterm delivery (Fig. 3g, Extended Data Fig. 9, Supplementary Table 3, and Methods). The performance for the specific preterm subtypes were 0.80 AUC (CI, 0.72–0.87; 71% sensitivity at 80% specificity) for EPE and 0.69 AUC (CI, 0.61–0.77; 50% sensitivity) for LPE-PB.

We validated the final model for predicting preeclampsia risk in a cohort of 831 routine PDNAS samples collected at a median of 12.2 weeks of gestation (PDNAS validation cohort; Fig. 1a,b). The overall performance for predicting preeclampsia requiring preterm delivery was 0.85 AUC (CI, 0.80–0.90; 81% sensitivity at 80% specificity), with AUCs of 0.85 (CI, 0.77–0.94; 82% sensitivity) and 0.84 (CI, 0.78–0.91; 80% sensitivity) for EPE and LPE-PB, respectively (Fig. 3h, Extended Data Fig. 9, and Supplementary Table 4). We applied the final model to an external cohort of 141 samples (median gestation, 12.1 weeks) collected across multiple biobanks and sequenced at an external clinical laboratory (external validation cohort; Fig. 1a,b, Supplementary Table 5, and Methods). The overall performance was 0.84 AUC (CI, 0.79–0.89; 79% sensitivity at 80% specificity) for predicting preeclampsia requiring preterm delivery, with 0.92 AUC (CI, 0.88–0.96; 94% sensitivity) for EPE and 0.79 (CI, 0.72–0.85; 68% sensitivity) for LPE-PB (Fig. 3i, Extended Data Fig. 9, and Supplementary Table 6). These results demonstrate that using the PEARL framework for cfDNA nucleosome accessibility analysis of early-pregnancy PDNAS data can stratify those at high risk of preeclampsia requiring preterm delivery, months before disease onset.

Discussion

Here we provide evidence that nucleosome profiling can be applied to clinically obtained low-coverage cfDNA sequence data to identify distinctive features among those who develop preeclampsia. We detected evidence of endothelial-tissue damage (placental and/or maternal) in early pregnancy, a major feature of preeclampsia pathogenesis^{30,31,38}, and decreased contributions from placental tissue. These findings informed PEARL for predicting preeclampsia risks months before the onset of clinical symptoms.

Currently available approaches, such as the Fetal Medicine Foundation first-trimester algorithm, outperform clinical risk factors alone (AUC 0.5 to 0.9)^{5,7}. This algorithm incorporates several maternal features, laboratory tests, and ultrasound measures (for example, racial categorization, levels of placental growth factor, and uterine artery pulsatility index)^{5,7}, but can be challenging to implement and is not translatable to all populations^{7,39}. Our approach, however, reliably predicts preeclampsia risk using cfDNA nucleosome profiling of placenta and endothelial tissue, with the addition of only two simple clinical parameters: blood pressure and BMI.

Screening for preeclampsia is most useful if effective interventions exist. Although the benefits of aspirin prophylaxis generally outweigh any risks^{40–43}, improved precision in risk stratification can eliminate unnecessary exposure. Seminal clinical trials indicate that stricter blood-pressure control can safely and effectively

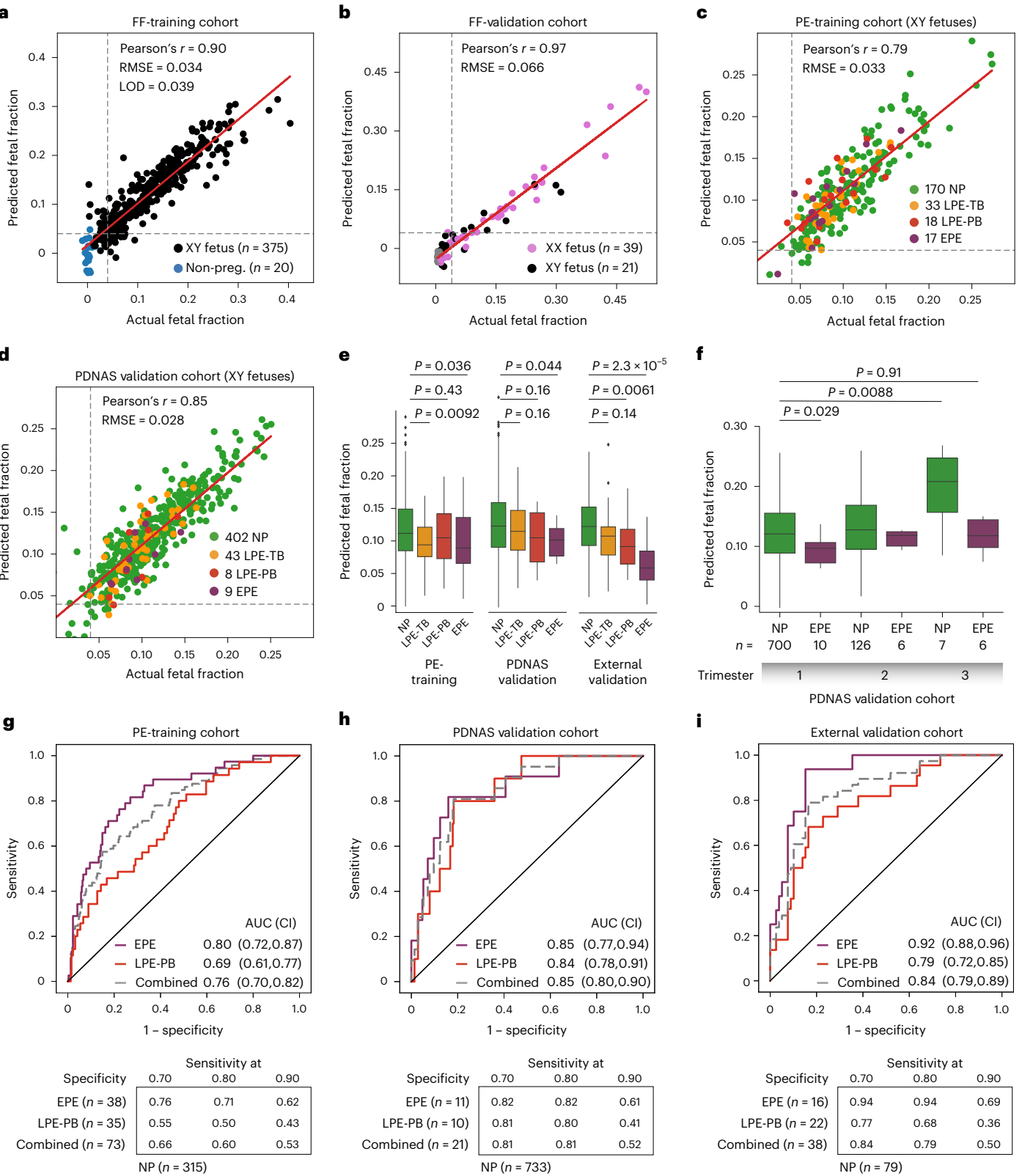
Fig. 3 | Nucleosome patterns from PDNAS estimate FF and predict PE risk.

a, PEARL-estimated FF in the FF-training cohort, which includes 375 PDNAS samples from XY fetuses (black dots) and 20 samples from non-pregnant females (blue dots). Performance was estimated using chromosome-Y-derived FF as the true FF in a bootstrapping with replacement framework with 1,000 iterations (Pearson's $r = 0.90$ (CI, 0.901–0.904); root mean square error (RMSE), 0.034 (CI, 0.033–0.033)). Using the maximum predicted FF in non-pregnancy samples, we estimated the limit of detection (LOD) at 0.039 (CI, 0.039–0.042), indicated by the horizontal dashed gray line. **b**, The final trained model was validated in the FF-validation cohort, in which the true FF had been determined using maternal–fetal genotyping. This cohort contains 39 samples from fetuses classified as female (pink dots), 21 as male (black dots), and 10 with unknown fetal sex (gray dots). **c**, PEARL-estimated FF on 238 XY samples from the PE-training cohort. **d**, PEARL-estimated FF on 462 XY samples from the PDNAS validation cohort. **e**, Median PEARL-estimated FF in the PE-training ($n = 450$), PDNAS validation ($n = 831$), and external ($n = 141$) cohorts. Samples from preterm deliveries with

PE (purple, EPE; red, LPE-PB) have a significantly lower PEARL-estimated FF compared to NP samples (green). **f**, PEARL-estimated FF in EPE compared to NP cases, stratified by trimester, from consecutively collected PDNAS samples at all available GAs in the PDNAS validation cohort. The PEARL-estimated FF remains lower across trimesters in cases which eventually developed EPE (purple). Boxes represent the IQR (25th–75th percentiles), with the horizontal line indicating the median. The whiskers extend to $1.5 \times$ IQR on either side, with dots representing outliers. Statistical analysis was done using the two-sided Mann–Whitney U test. **g–i**, Receiver operating characteristic curves for predicting risk for developing PE with preterm delivery, using samples collected at ≤ 16 weeks GA (combined EPE and LPE-PB, gray dashed line) from the PE-training cohort ($n = 450$) (**g**), PDNAS validation cohort ($n = 831$) (**h**), and external cohort ($n = 141$) (**i**). Samples are further stratified as EPE (purple) and LPE-PB (red). Sensitivity is listed (bottom) for three set specificities. Performance metrics calculated for sensitivity and AUC (95% CI) were generated using a bootstrapping with replacement framework with 1000 iterations. The Mann–Whitney U test was used for pairwise comparisons.

decrease the incidence of severe hypertension, preeclampsia, and hypertension-associated preterm birth^{44–46}. False positive predictions through a potential screening approach in this clinical setting generally carry low risks as these interventions have established safely. The favorable performance of our prediction model could facilitate stratification of risk during early pregnancy to test these interventions.

Although recent methods that utilize plasma-derived cfRNA and cfDNA can also facilitate preeclampsia prediction^{15–17}, these approaches require specialized sequencing assays that are challenging to implement, even in high-income settings. Given the success of aneuploidy screening in prenatal medicine, our method is advantageous because it uses an existing clinical assay without requiring additional sampling or sequencing. Because this approach still requires transporting



samples to sequencing facilities, which are typically available only in high-income settings²⁰, further innovations are needed to facilitate implementation in low- and middle income countries.

Prior studies using PDNAS cfDNA sequence data either did not investigate adverse pregnancy outcomes or were not generalized to an unselected population^{47,48}. By contrast, we focused on preeclampsia risk prediction, demonstrating favorable performance in EPE and LPE-PB cases, with modest performance in LPE-TB cases. Furthermore, we analyzed consecutively obtained PDNAS samples with broad inclusion criteria, approximating true preeclampsia prevalence and patient variability. Importantly, the PDNAS validation cohort did not exclude patients with pre-existing hypertension, diabetes, or other maternal medical conditions and included pregnancies ending in preterm birth that were not complicated by preeclampsia, closely approximating a real-world cohort and strengthening the validity of our findings of cfDNA tissue signatures specific to preeclampsia. This broad, inclusive approach, coupled with our favorable results, sets the stage for larger studies to validate the clinical utility of this method in a prospective manner in which pregnancy outcomes are unknown.

We demonstrate that nucleosome profiling of cfDNA sequence data derived from PDNAS can accurately estimate FF, overcoming some current limitations, and can be expanded to include risk stratification for the development of preeclampsia requiring preterm delivery, a current unmet need. Importantly, our approach can be applied to low-coverage standard sequencing data, highlighting the potential for scalability, cost-effectiveness, and lower barriers for clinical implementation.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41591-025-03509-w>.

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Methods

Ethics and inclusion statement

Institutional Review Board approval was obtained from UW to use PDNAS sequence data housed at our institution (STUDY0001636, STUDY00005540, STUDY00014943) and samples obtained from biorepositories for the external validation cohort (STUDY00004878, STUDY00017462). Informed consent was obtained as required by UW Institutional Review Board policies, or per the policies at the external biorepository sites. PDNAS samples for our study were primarily derived from individuals seeking prenatal care and/or genetic screening through our institution, which is a tertiary referral center covering a five-state region. The external validation cohort sequenced elsewhere included samples with patient demographics different from those of our cohort. We recognize the lack of geographic representation of low- and middle-income countries (LMICs) in this proof-of-concept study. We acknowledge that LMIC representation in large prospective studies is crucial to determine whether our findings are widely applicable and can benefit all pregnant people.

Classification of normal pregnancy

We designated a pregnancy as normal if it was a singleton gestation; continued past 24 weeks; was not complicated by preeclampsia (including postpartum preeclampsia), gestational hypertension, maternal transplant, or maternal malignancy; and did not have aneuploidy or a major fetal anomaly. In the PE-training cohort ($n = 450$), we included only those with pregnancies ending in a term delivery (≥ 37 weeks) in the normal group. In the PDNAS validation cohort ($n = 831$), however, we included cases with preterm delivery (< 37 weeks) in the normal group. To assemble cohorts more representative of a real-world clinical population, we did not exclude individuals with pre-existing hypertension, diabetes, or other maternal medical conditions (for example, renal disease or systemic lupus erythematosus) from any of the three cohorts (PE-training, PDNAS validation, or external).

Clinical diagnosis of preeclampsia

Preeclampsia diagnosis was based on the American College of Obstetricians and Gynecologists criteria in pregnancies continuing past 20 weeks gestation¹. We excluded non-singleton gestations, cases of chronic hypertension without superimposed preeclampsia (retained in the normal cohort), postpartum preeclampsia, gestational hypertension, maternal transplant, maternal malignancy, or if the pregnancy was complicated by aneuploidy or major fetal anomaly. Preeclampsia was designated as EPE if diagnosis occurred at or before 34 weeks gestation and delivery was preterm (< 37 weeks); or as LPE if confirmed after 34 weeks gestation. For clinical significance, LPE was further classified as LPE-PB or LPE-TB, using a gestational age at delivery of < 37 weeks to define preterm birth. Medical records were reviewed by trained maternal–fetal medicine providers (T.K. and R.S.) to confirm the preeclampsia diagnosis. For preeclampsia prediction, we primarily focused our results on EPE and LPE-PB, because these cases required preterm deliveries, which are associated with a high risk of neonatal morbidity and mortality⁴⁹. All samples are from individuals of female sex based on the designation in the electronic health record. No information on gender identity was collected.

Training and validation cohorts

Since May 2017, UW has clinically used an internally developed PDNAS assay that employs low-pass WGS of cfDNA extracted from maternal plasma to provide a high-quality screen for autosomal and sex-chromosome aneuploidies. A total of 1,854 clinical PDNAS samples were included in this study; samples were divided as outlined in Fig. 1. The FF-training cohort consists of 375 PDNAS samples with XY fetuses and 20 from healthy non-pregnant females sequenced using the same clinical assay (total $n = 395$). The FF-training cohort was used to train the PEARL-estimated FF model. The PE-training cohort was assembled by

randomly selecting approximately 10% of samples that met inclusion criteria and were clinically obtained from 1 January 2019 to 31 December 2020, with enrichment of preeclampsia cases from samples clinically obtained from 1 January 2021 to 25 April 2023. From here, we excluded 54 samples that were collected after 16 weeks gestation (total $n = 450$). The PDNAS validation cohort initially consisted of 956 samples and resembles an all-comers population because it includes all consecutively obtained clinical samples from 1 May 2017 to 31 December 2018 that met the broad inclusion criteria as detailed above. Notably, the normal group in this cohort included all term and preterm deliveries not complicated by preeclampsia. After excluding 125 samples that were collected after 16 weeks gestation, the final PDNAS validation cohort used for preeclampsia risk prediction consisted of 831 samples. Samples used for training (FF-training cohort for FF estimation and PE-training cohort for preeclampsia risk prediction) were distinct and separate from those used for validation of each of these models.

FF external validation cohort

For external validation of the PEARL-estimated FF model, we used published data with single-nucleotide polymorphism (SNP)-based quantification of FF from pregnancies with male and female fetuses ($n = 70$)³⁷. These samples were sequenced to a median depth of $4\times$, enabling down-sampling analysis to assess the effect of sequence depth on FF determination. Down-sampling was performed at depths of $0.1\times$, $0.5\times$, $1\times$, and $1.5\times$. Sequencing read duplicates were first marked on the aligned bam file using Picard Tools (v2.18.29, <http://broadinstitute.github.io/picard>) MarkDuplicates and down-sampling was performed using DownsamplingSam.

External validation cohort for preeclampsia risk prediction

For external validation of the preeclampsia risk prediction model, we assembled 146 samples (stored plasma or extracted cfDNA) from two biobanks (the Magee Obstetrical Maternal Infant Database and Biobank and the UW Pregnancy Research Biobank) enriched for preeclampsia cases ($n = 16$ EPE, $n = 23$ LPE-PB, $n = 23$ LPE-TB, and $n = 79$ normal controls). These samples were sequenced at an external clinical laboratory (the Broad Institute), which is accredited by the Clinical Laboratory Improvement Amendments and The College of American Pathologists (CLIA/CAP), at a low-pass sequence depth (mean coverage, $2.3\times$; IQR, 1.19 – 3.11).

University of Washington PDNAS library protocol

For PDNAS samples sequenced through the UW clinical laboratory, libraries were generated using a validated, laboratory-developed method that has been previously reported^{50,51}. In brief, whole blood from Streck (BCT1) tubes was centrifuged, and plasma was isolated following the instructions on the package insert. cfDNA was extracted using the QIAasympyphony Circulating DNA Kit (Qiagen). Following DNA quantification, next-generation sequencing library preparation was performed using the KAPA HyperPrep kit for adapter and index ligation. Libraries were purified before and after amplification using the AMPureXP kit (Beckman Coulter). Sample pools were created using an equimolar strategy and diluted to 1 nM. Sequencing was performed using an Illumina NextSeq 500 with a 37-bp paired-end read configuration.

Sequence data processing

All sequence data (PDNAS samples and external validation samples) were aligned to the reference genome hg38 (GRCh38, https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000001405.26/). Raw fastq files were adapter-trimmed using the cutadapt tool (v3.3)⁵². The paired reads were then aligned using BWA-MEM algorithm (v0.7.17) and were further processed using SAMtools (v1.10) to sort and index the resulting bam file^{53,54}. Picard tool's (v2.18.29, <http://broadinstitute.github.io/picard>) MarkDuplicates function was used to mark duplicates. Picard tool's CollectAlignmentSummaryMetrics, CollectWgsMetrics and

CollectInsertSizeMetrics were used to calculate quality-control (QC) sequencing metrics.

Chromosome-Y-based fetal fraction calculation

The UW clinical lab uses the following procedures for determining the FF for PDNAS samples from pregnancies with male fetuses. The chromosome-Y-based FF was calculated on the basis of the percentage of reads that map to chromosome Y (chrY) using the formula:

$$\text{ChrY}\% = \frac{\Sigma(\text{Reads mapped to ChromosomeY})}{\Sigma(\text{All reads})}$$

To adjust for erroneously mapped reads to chrY, a baseline set of male samples and samples with known female fetuses were used to estimate the average percentage of reads mapping to chromosome Y:

$$\text{Average male ChrY}\% = \Sigma\{\text{Male ChrY}\%\}/n$$

$$\text{Average female ChrY}\% = \Sigma\{\text{Female ChrY}\%\}/n$$

Average male ChrY% is the percentage of reads mapped to chromosome Y from male donors. For average female ChrY%, samples with known female fetuses were used. Finally, the FF is calculated by adjusting for the average erroneously mapped reads:

$$\text{ChrY} - \text{FF} = \frac{(\text{ChrY}\% - \text{Average female ChrY}\%)}{(\text{Average male ChrY}\% - \text{Average female ChrY}\%)}$$

PEARL generating nucleosome-based tissue profiles from cfDNA

Curating tissue-specific and transcription-factor binding sites. Tissue-specific DNase I hypersensitivity sites (DHSs) were obtained from the ENCODE regulatory index⁵⁵ to construct 16 tissue-specific open chromatin maps (<https://www.meuleman.org/research/dhsindex/>). The top 10,000 DHS summits from each tissue were selected on the basis of ranking by highest recurrence across experiments ('numsamples') and mean DHS signal ('mean_signal'). Additionally, 19 tissue-specific open chromatin maps were generated from published single-cell transposase-accessible chromatin sequencing (scATAC-seq) data (<http://catlas.org/humanenhancer>)⁵⁶. For open chromatin sites, the top 10,000 peaks were selected by peak scores. Transcription-factor-binding sites (TFBSs) for 377 transcription factors were obtained from the Gene Transcription Regulation Database (GTRD)⁵⁷, and the top 10,000 TFBSs were filtered on the basis of recurrence across experiments, as described in our previous study²².

Generating coverage profiles using Griffin. The PEARL framework starts with the generation of nucleosome coverage profiles using Griffin v0.2.0 (<https://github.com/GavinHaLab/Griffin/releases/tag/v0.2.0>) to compute a composite coverage profile, aggregating across sets of sites for each tissue and cell type or sets of TFBSs for each transcription factor²². Coverage profiles were generated for a ± 2 -kb window flanking the peak summit (for open chromatin) or binding site (for TFBS). Coverage profiles were further restricted to use specific cfDNA reads on the basis of fragment sizes: nucleosome profile (120–180 bp) and subnucleosomal profile (35–80 bp).

Computing NDR and MCV scores. From the coverage profiles, two features were extracted: the NDR, defined as the mean Griffin-normalized coverage value ± 30 bp and ± 75 bp from the summit or binding site, and MCV, defined as the mean Griffin-normalized coverage value $\pm 1,000$ bp from the summit or binding site. For single nucleosome profiles, lower values correspond to increased accessibility and increased tissue contribution. For sub-nucleosome profiles, higher values correspond

to increased accessibility and increased tissue contribution. These features are used for tissue of origin, FF estimation, and PE risk prediction in the PEARL framework.

PEARL tissue-of-origin analysis

We investigated overall tissue contributions in maternal plasma in 375 PDNAS samples from pregnant individuals in the FF-training cohort. We then investigated tissue contributions across gestation in the FF-training cohort, comprising 237 first trimester (≤ 14 weeks gestation), 122 second trimester (14–28 weeks gestation), and 16 third trimester (> 28 weeks gestation) samples, compared to 20 non-pregnant female samples. We also performed Pearson's r correlation analysis between gestational age at time of PDNAS for each tissue-specific feature (NDR and MCV) from sites derived from both scATAC-seq and DNase-seq. Lastly, we used Pearson's r correlation analysis to evaluate the effect of BMI on tissue-specific features.

To investigate placental and endothelial tissue contributions in early-pregnancy plasma that are associated with preeclampsia, we compared PDNAS samples from the PE-training cohort collected at ≤ 16 weeks gestation from normal pregnancies ($n = 315$) with 135 preeclampsia cases (38 EPE, 35 LPE-PB, and 62 LPE-TB). We performed the Mann–Whitney U test (scipy.stats.mannwhitneyu, v1.11.0) with Bonferroni correction ($\alpha = 0.05$, statsmodels.stats.multitest.multipletests, v0.13.2). To account for BMI and FF, which are known to influence PDNAS test characteristics^{51,58}, we also performed analysis of covariance (ANCOVA) using pingouin (v0.5.3) by including these as covariates⁵⁹.

PEARL FF determination on the basis of nucleosome profiling

We used 395 PDNAS samples (FF-training cohort) consisting of 375 XY fetuses and 20 non-pregnant healthy females to train a model for FF estimation using our nucleosome profiling method, termed PEARL-estimated FF. Feature selection, training, and testing were performed in a cross-validation framework using bootstrapping with replacement of 1,000 iterations. For each bootstrap, samples were sampled with replacement, and the remaining samples were included in the hold-out set for that iteration. The top placental specific features were identified from the training set by analyzing 16 tissue-specific DHSs, 20 tissue-specific scATAC sites, and 377 TFBS nucleosome and sub-nucleosome features (nucleosome, MCV and NDR; sub-nucleosome, NDR; total, 1,652) between samples from non-pregnant controls ($n = 20$) and pregnancy samples with a high (> 0.25) chromosome Y FF (chrY-FF) using the Mann–Whitney U test (scipy.stats.mannwhitneyu, v1.11.0) with Bonferroni correction ($\alpha = 0.05$, statsmodels.stats.multitest.multipletests, v0.13.2). Statistically significant ($P < 0.05$) features were used in a supervised Bayesian Ridge regression framework (sklearn.BayesianRidge, v1.1.1) to train a linear model for estimating the FF as a continuous value, using chrY-FF value as the true value. In each iteration, the trained model was applied to the hold-out set, and the RMSE and Pearson's r were calculated. The model was also applied to the non-pregnant hold-out samples, and the limit of detection (LOD) was calculated as the maximum predicted chrY-FF from any non-pregnancy sample. The overall training performance was reported as the average for the RMSE, Pearson's r , and LOD across 1,000 iterations with 95% CIs.

Final PEARL model for FF estimation. The final model for estimating FF using PEARL was then trained on all 395 PDNAS samples. First, using all 1,239 features, 141 features were identified from all non-pregnancy samples ($n = 20$) and pregnancy samples with a high chrY-FF (> 0.25) ($n = 25$). A supervised Bayesian Ridge regression model was then trained using the 141 features across all 395 samples. The final model was then applied to a separate external validation cohort (FF-validation cohort, $n = 70$) and all preeclampsia cohorts (PE-training cohort and PDNAS validation cohort). Performance metrics were calculated similarly to above, using bootstrapping with replacement across 1,000 iterations.

PEARL validation of FF derived from nucleosome profiling

We accessed data for 70 pregnancy samples from an external publicly available dataset (Jiang et al.)³⁷, in which the FF was confirmed using maternal–fetal SNP genotyping. Fetal sex was predicted using the chromosome-Y ratio for samples with FF > 0.10 (39 female fetuses, 21 male fetuses, and 10 with unknown sex). These samples were sequenced to a median depth of 4×, allowing for down-sampling analysis to compare the effect of sequence depth on FF determination using PEARL. Nucleosome-based tissue profiles were then generated for all tissue and cell types and TFBSs using Griffin. NDR and MCV scores were generated as described above. The final trained model (trained on the 395 samples in the FF-training cohort) was applied to these externally sequenced samples and performance metrics, such as RMSE and Pearson's *r*, were computed.

PEARL development of preeclampsia risk prediction model

Patient cohort and model. A total of 450 PDNAS samples (collected ≤16 weeks gestation, with confirmed clinical outcome) were part of the PE-training cohort, which included 135 cases of preeclampsia and 315 normal pregnancies (Fig. 1a,b). We considered three feature sets defined as follows:

- (1) Placental- and endothelial-tissue-specific chromatin profiles (nucleosome, MCV and NDR; sub-nucleosome, NDR) and FF as estimated by PEARL.
- (2) Systolic blood pressure, diastolic blood pressure and BMI.
- (3) Combination of sets (1) and (2).

We included only BMI and blood pressure as clinical parameters because they can be easily ascertained to minimize clinician burden, especially because some conditions (for example, systemic lupus erythematosus) require complex diagnostic confirmation. Furthermore, we aimed to identify markers in cfDNA that demonstrate favorable performance without the need for multiple clinical inputs.

Machine-learning prediction model and training. Samples were trained and tested using bootstrapping with replacement ($n = 1,000$). Three models were trained for each feature set, l1 logistic regression, l2 logistic regression (sklearn LogisticRegression with class weight balanced, v1.1.1), and XGBoost (xgboost XGBClassifier, number of estimators = 10 and max depth = 5, v1.6.1). These models were then used to form a soft ensemble model (mean probability across three models). In each iteration, the model is trained as a supervised binary classifier using all normal and preeclampsia samples from the respective training cohort.

Training performance. The trained model was applied to hold-out test samples, and the final predicted probability is taken as the mean of the three classifiers for each feature set. Through each iteration of bootstrapping, the following was computed: AUC, false positive rate (FPR), true positive rate (TPR), and sensitivity for a set specificity of 0.7, 0.8, and 0.9. A positive case is defined as an individual with preeclampsia (EPE, LPE-PB, or LPE-TB), and a negative case is an individual with normal pregnancy.

Final preeclampsia model in PEARL. The final model for predicting risk for preeclampsia was then trained using all 450 PDNAS samples in the PE-training cohort. This final trained model was then applied to the PDNAS validation cohort ($n = 831$ samples) and the multicenter external cohort ($n = 141$ samples). Performance metrics were calculated similarly to above, using bootstrapping with replacement across 1,000 iterations.

PEARL validation of the preeclampsia risk prediction model

Internal validation of the preeclampsia prediction model was performed on 831 samples (PDNAS validation cohort), distinct from those used to train the model (PE-training cohort). The study team (except for T.K. and

R.S., who adjudicated cases) was blinded to the clinical outcome (normal versus preeclampsia). External validation involved application of the PEARL preeclampsia risk prediction model on sequencing results from 141 samples assembled from two biobanks and sequenced at an external clinical laboratory. Performance metrics (AUC, FPR, TPR, and sensitivity for a set specificity of 0.7, 0.8, and 0.9) of the PEARL prediction model on the internal and external validation cohorts were calculated.

Statistical analysis

Sequenced data with available clinical outcome information were considered for inclusion: PDNAS validation cohort, consecutively obtained clinical samples from 1 May 2017 to 31 December 2018; PE-training cohort, random selection of approximately 10% of samples collected from 1 January 2019 to 31 December 2020. Samples were excluded on the basis of criteria listed in 'Classification of normal pregnancy' and 'Clinical diagnosis of preeclampsia' sections. No statistical method was used to predetermine sample size.

All two-group comparisons of continuous values were performed using the Mann–Whitney *U* test; *P* values were estimated using the default two-sided test, with the statistical significance level set at 0.05. Pearson's correlation was used to compare two continuous variables; the correlation coefficient, *r*, is reported, and the *P* value was estimated using the default two-sided test with the CI set at 95%. The RMSE was used to determine the error as the average difference between the model-predicted FF and the actual FF. Receiver operating characteristic curves were used to evaluate classification performance; the AUC was used as the metric for performance. Confidence intervals (95%) for AUCs were computed using 1,000 iterations of bootstrapping with replacement. True positive (TP) was defined as correctly predicting preeclampsia; false positive (FP) was defined as predicting preeclampsia for a normal pregnancy; true negative (TN) was defined as correctly predicting normal pregnancy; and false negative (FN) was defined as predicting normal pregnancy for a preeclampsia case. Sensitivity was computed as $TP / (TP + FN)$; specificity was computed as $TN / (TN + FP)$.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data necessary to reproduce the findings in this study can be accessed at <https://github.com/GavinHaLab/PEARL/> and the Supplementary Tables. Cell-free DNA sequence data are housed at the University of Washington; given the privacy restrictions, reuse of these genomic data may be possible upon reasonable request pending approval by the University of Washington Agreements Office, including execution of a Data Transfer and Use Agreement. Inquiries for data usage can be directed to R.S. (shreer23@uw.edu), and a response can be expected within two weeks. Sequence data were aligned to reference genome hg38 (GRCh38, https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000001405.26/). Tissue-specific binding sites were derived from the following accessibility datasets: <https://www.meuleman.org/research/dhsindex/>, <http://catlas.org/humanenhancer>. Source data are provided with this paper.

Code availability

Custom code, scripts, and models for the PEARL framework can be accessed at <https://github.com/GavinHaLab/PEARL/>. The Griffin software used in this study can be accessed at <https://github.com/GavinHaLab/Griffin/releases/tag/v0.2.0>.

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Author contributions

Conceptualization: M.A., J.B.R., C.M.L., G.H., R.S. Methodology: M.A., J.B.R., G.H. Software: M.A., A.-L.D., G.H. Formal Analysis: M.A., J.B.R., T.R.K., A.-L.D., E.K., B.C., M.Y., R.D.P., T.W.P., G.H., R.S. Investigation: M.A., J.B.R., T.R.K., A.-L.D., G.H., R.S. Resources: C.C.P., C.M.L., R.S. Data Curation: T.R.K., L.A.C., C.L.T., P.G., S.R., C.M.L., R.S. Writing – Original Draft: M.A., G.H., R.S. Writing – Review and Editing: M.A., T.R.K., J.B.R., A.-L.D., L.A.C., P.G., B.C., R.D.P., C.C.P., S.A., C.M.L., G.H., R.S. Visualization: M.A., G.H., R.S. Supervision: G.H., R.S. Funding Acquisition: C.C.P., G.H., R.S.

Competing interests

M.A., G.H., J.B.R., C.M.L., and R.S. are inventors on patent applications relating to the methodologies developed and applied in this paper. C.M.L. is on the Scientific Advisory Board for LGC Group which sells reference material samples for clinical labs. The other authors declare no competing interests.

Additional information

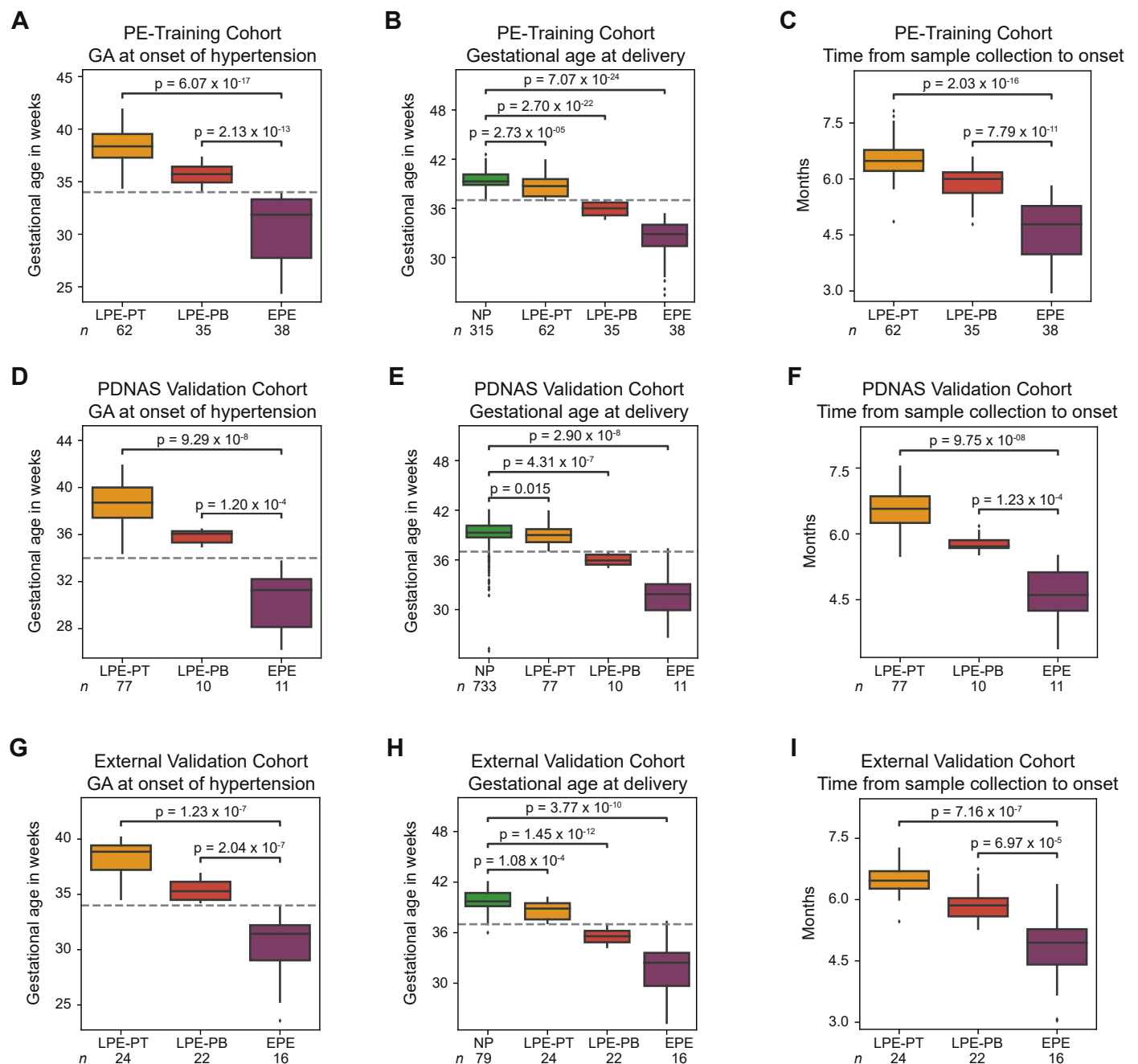
Extended data is available for this paper at <https://doi.org/10.1038/s41591-025-03509-w>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41591-025-03509-w>.

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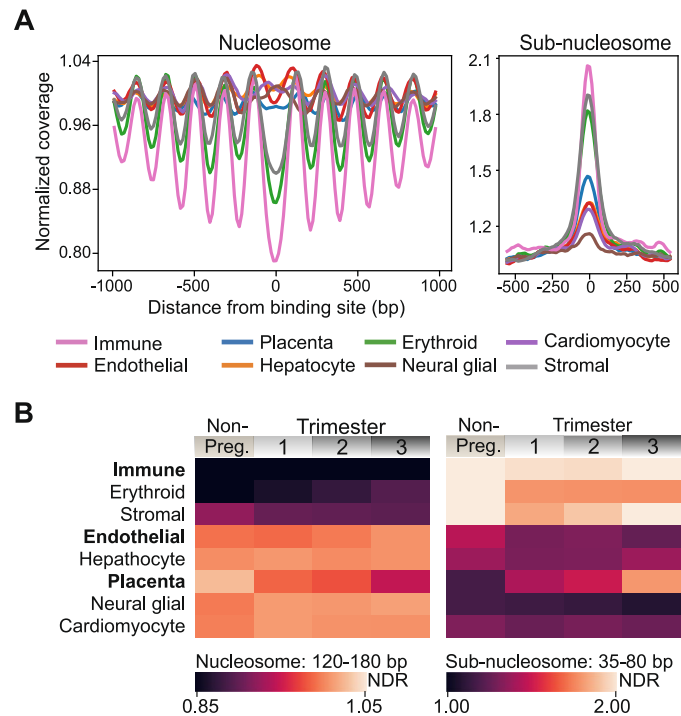
Peer review information *Nature Medicine* thanks Annie McDougall and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Sonia Muliyl, in collaboration with the *Nature Medicine* team.

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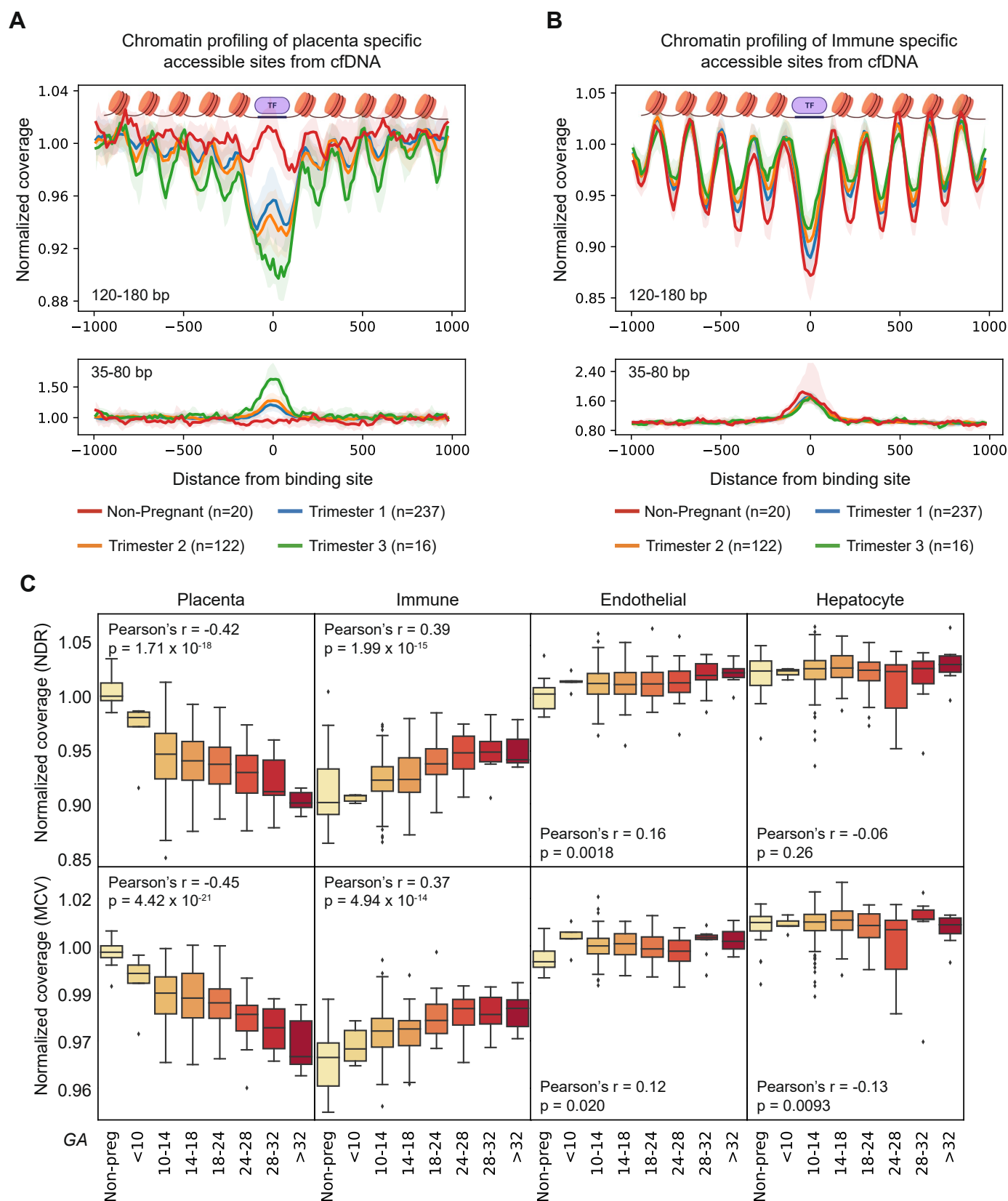
Extended Data Fig. 1 | Gestational age at onset of hypertension, gestational age at delivery, and interval from sample collection to onset of hypertension in each of the three cohorts. (a) GA at onset of hypertension for PE cases in the PE-Training Cohort. Note cases with onset of hypertension before 34 weeks were designated as early preeclampsia (EPE). The remaining cases were further classified as either late preeclampsia with preterm birth (LPE-PB) or late preeclampsia with term birth (LPE-TB). **(b)** GA at delivery for each outcome in the PE-Training Cohort. Pregnancies with delivery before 37 weeks (dashed grey line) were considered to have a preterm birth. **(c)** Interval from sample collection to clinical onset of hypertension in the PE-Training Cohort. Note that the onset of hypertension for EPE is at a median of 4.79 months after PDNAS. **(d)** GA at onset of hypertension for PE cases in the PDNAS Validation Cohort. Note cases with onset of hypertension before 34 weeks were designated as EPE. The remaining cases were further classified as either LPE-PB or LPE-TB. **(e)** GA at delivery for each

outcome in the PDNAS Validation Cohort. Pregnancies with delivery before 37 weeks (dashed grey line) were considered to have a preterm birth. **(f)** Interval from sample collection to clinical onset of hypertension in the PDNAS Validation. Note that the onset of hypertension for EPE is at a median of 4.61 months after PDNAS. **(g)** GA at onset of hypertension for PE cases in the External Validation Cohort. **(h)** GA at delivery for each outcome in the External Validation Cohort. Pregnancies with delivery before 37 weeks (dashed grey line) were considered to have a preterm birth. **(i)** Interval from sample collection to clinical onset of hypertension in the External Validation Cohort. Note that the onset of hypertension for EPE is at a median of 4.95 months after PDNAS. Statistical comparison was done using two-sided Mann-Whitney U test. PDNAS, prenatal cell-free DNA (cfDNA) screening; PE, preeclampsia; GA, gestational age; NP, normal pregnancy; EPE, early preeclampsia; LPE-PB, late preeclampsia with preterm birth; LPE-TB, late preeclampsia with term birth.



Extended Data Fig. 2 | Discernment of distinct tissue types overall and by trimester using nucleosome profiling. (a) Nucleosome profile (left) and sub-nucleosome profile (right) of tissue specific accessible sites derived from DNase-seq in 395 PDNAS samples (FF-Training Cohort). Note that immune (pink) and placenta (blue) have lower and higher NDR values in nucleosome and sub-nucleosome profiles respectively, indicating higher tissue contribution.

(b) Heatmap of normalized coverage values at NDR for tissue specific nucleosome (left) and sub-nucleosome (right) features derived from DNase-seq for 20 non-pregnant and 375 pregnant PDNAS samples stratified by trimester. PDNAS, prenatal cell-free DNA (cfDNA) screening; PE, preeclampsia; GA, gestational age; Non-Preg, non-pregnant samples; NDR, nucleosome depleted region.

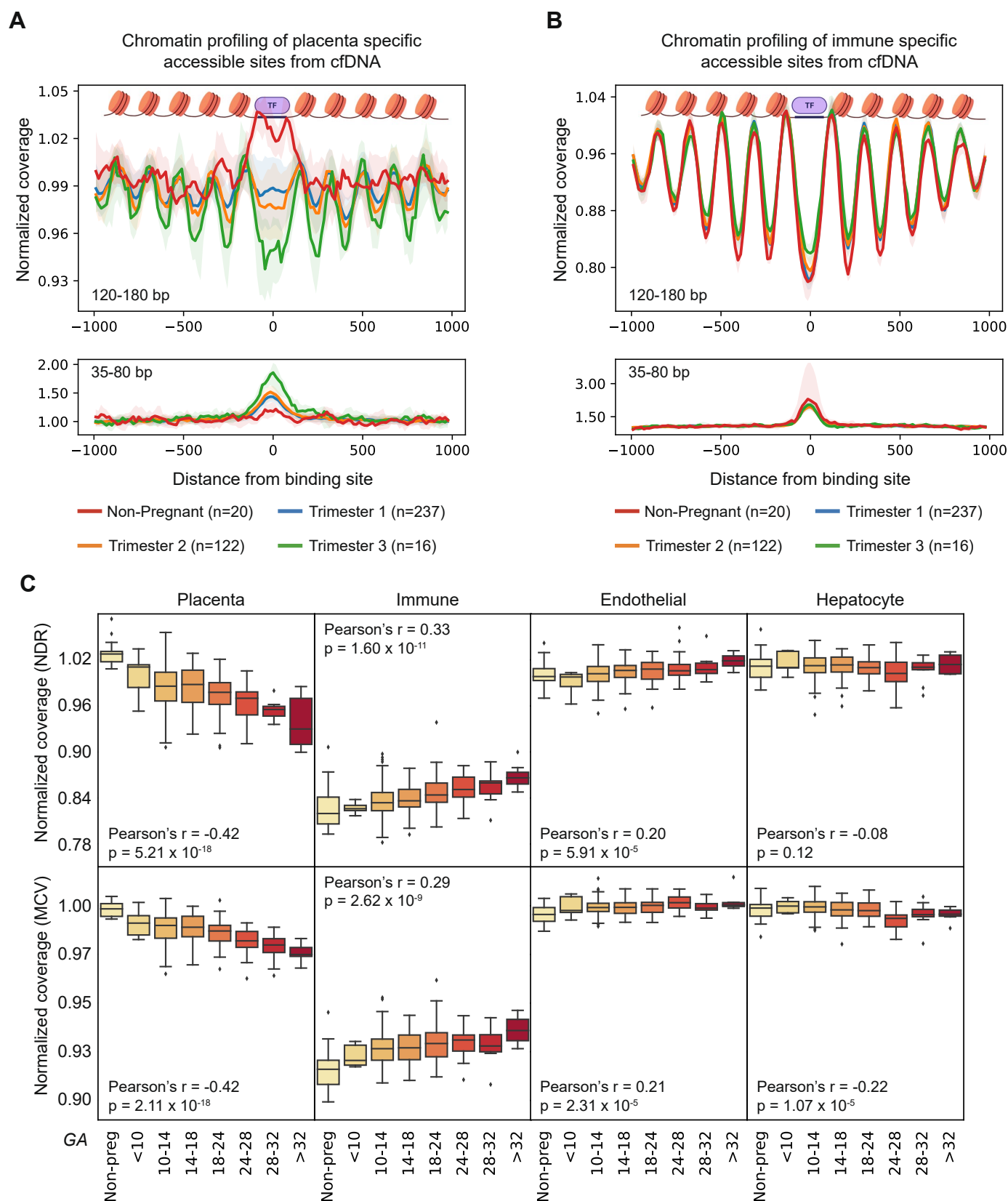


Extended Data Fig. 3 | See next page for caption.

Extended Data Fig. 3 | Nucleosome profiles for placental- and immune-specific accessible sites by trimester, derived from scATAC-seq dataset.

(a) Nucleosome profile for placenta specific accessible sites derived from scATAC-seq dataset for 20 non-pregnant females and 375 PDNAS samples from FF-Training Cohort stratified by trimester at sampling. Note decrease in nucleosome coverage (120–180 bp) indicates increased placental tissue contribution. Increase in sub-nucleosome coverage (35–80 bp) indicates increased tissue contribution. **(b)** Nucleosome profile for immune specific accessible sites derived from scATAC-seq dataset for 20 non-pregnant females and 375 PDNAS samples from FF-Training Cohort stratified by trimester at sampling. Note decrease in nucleosome coverage (120–180 bp) indicates increased placental tissue contribution. Increase in sub-nucleosome coverage

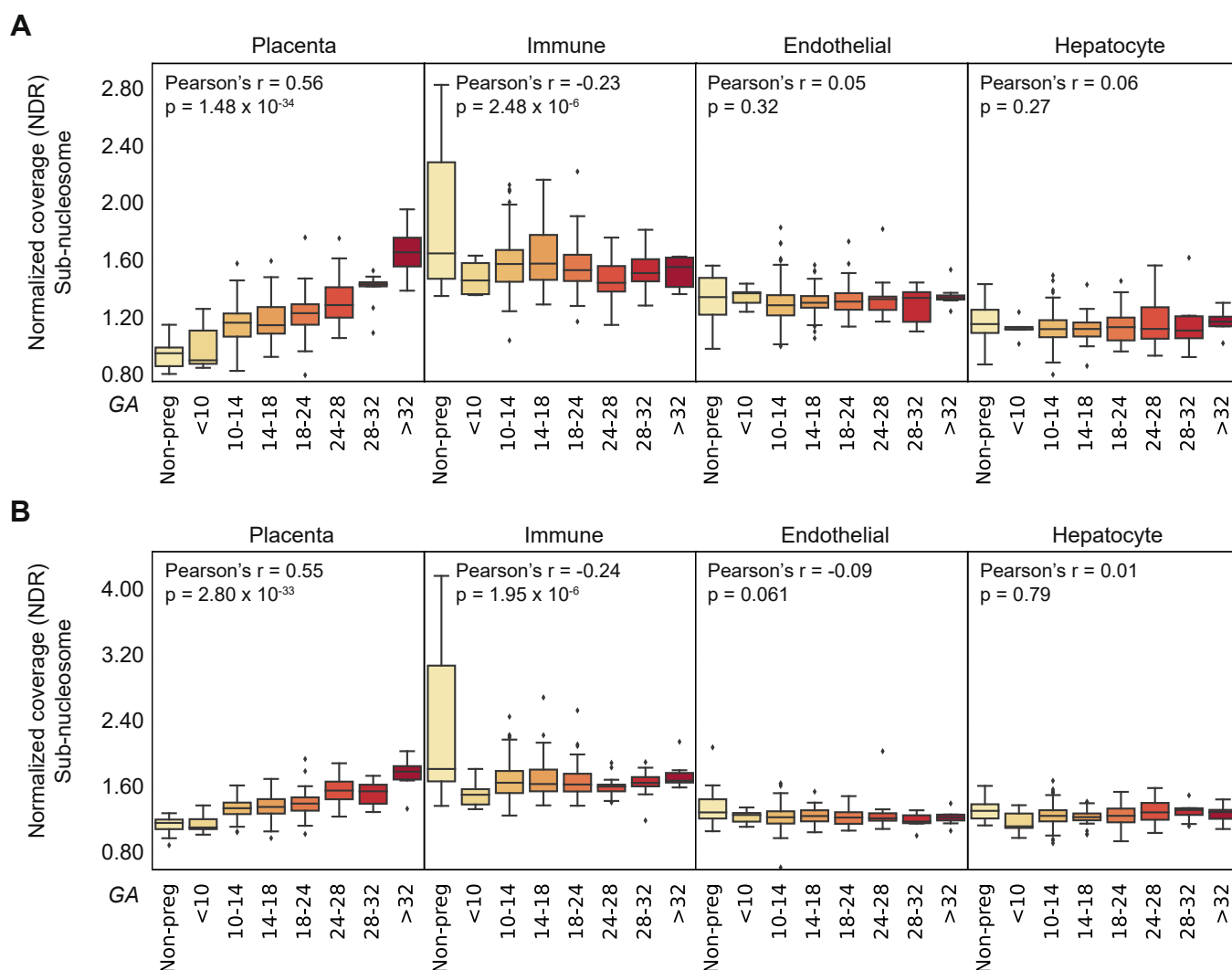
(35–80 bp) indicates increased tissue contribution. Shading represents the interquartile range (IQR, 25th and 75th percentiles) with bold line representing median. **(c)** Boxplot of nucleosome depleted regions (NDR) and mean central coverage (MCV) values at tissue specific accessible sites derived from scATAC-seq for 20 non-pregnant and 375 PDNAS samples from FF-Training Cohort stratified by GA in weeks of plasma collection. Correlation of tissue values is measured using Pearson's r . The box represents the interquartile range (IQR, 25th and 75th percentiles) with the horizontal line indicating median. The whiskers extend to 1.5X the IQR on either side, with dots representing outliers. *PDNAS*, prenatal cell-free DNA (*cfDNA*) screening; *GA*, gestational age; *NDR*, nucleosome depleted region; *MCV* mean central coverage.



Extended Data Fig. 4 | See next page for caption.

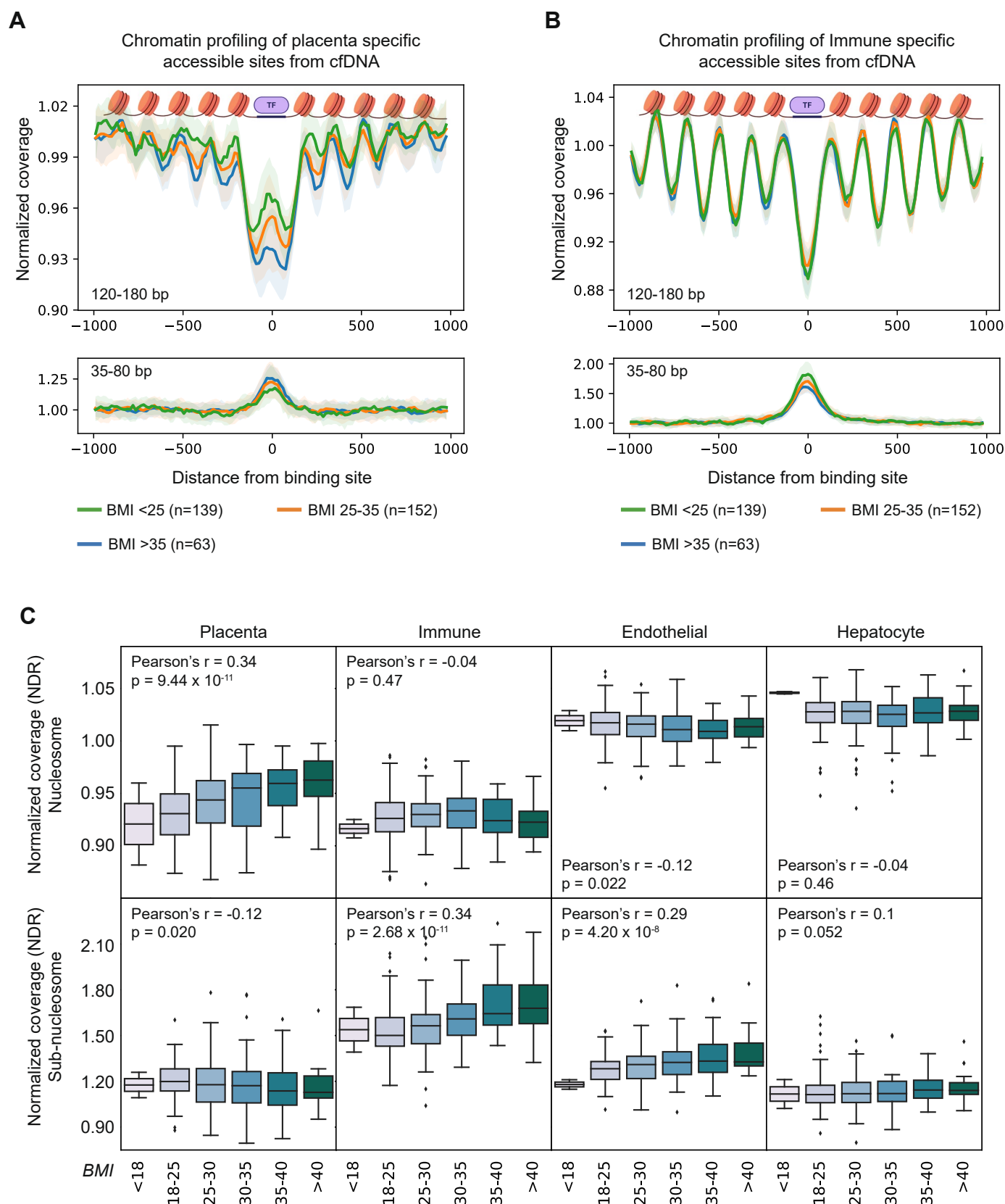
Extended Data Fig. 4 | Nucleosome profiles for placental- and immune-specific accessible sites by trimester, derived from DNase-seq dataset. (a) Nucleosome profile for placenta specific accessible sites derived from DNase-seq dataset for 20 non-pregnant females and 375 PDNAS samples from FF-Training cohort stratified by trimester at sampling. Note decrease in nucleosome coverage (120–180 bp) indicates increased placental tissue contribution. Increase in sub-nucleosome coverage (35–80 bp) indicates increased tissue contribution. (b) Nucleosome profile for immune specific accessible sites derived from DNase-seq dataset for 20 non-pregnant females and 375 PDNAS samples from FF-Training cohort stratified by trimester at sampling plasma. Note decrease in nucleosome coverage (120–180 bp) indicates increased placental tissue contribution. Increase in sub-nucleosome coverage (35–80 bp) indicates

increased tissue contribution. Shading represents the interquartile range (IQR, 25th and 75th percentiles) with bold line representing median. (c) Boxplot of nucleosome depleted regions (NDR) and mean central coverage (MCV) values at tissue specific accessible sites derived from DNase-seq for 20 non-pregnant and 375 PDNAS samples from FF-Training cohort stratified by GA in weeks of plasma collection. Correlation of tissue values is measured using Pearson's r . The box represents the interquartile range (IQR, 25th and 75th percentiles) with the horizontal line indicating median. The whiskers extend to 1.5X the IQR on either side, with dots representing outliers. *PDNAS*, prenatal cell-free DNA (*cfDNA*) screening; *GA*, gestational age; *NDR*, nucleosome depleted region; *MCV* mean central coverage.



Extended Data Fig. 5 | Nucleosome-depleted regions of subnucleosome profile at tissue specific accessible sites across gestation from scATAC-seq and DNA-seq datasets. (a) Boxplot of nucleosome depleted regions (NDR) of subnucleosome profile at tissue specific accessible sites derived from scATAC-seq for 20 non-pregnant and 375 PDNAS samples from FF-Training cohort stratified by GA in weeks of plasma collection. Correlation of tissue values is measured using Pearson's r . **(b)** Boxplot of nucleosome depleted regions (NDR) of subnucleosome profile at tissue specific accessible sites derived from

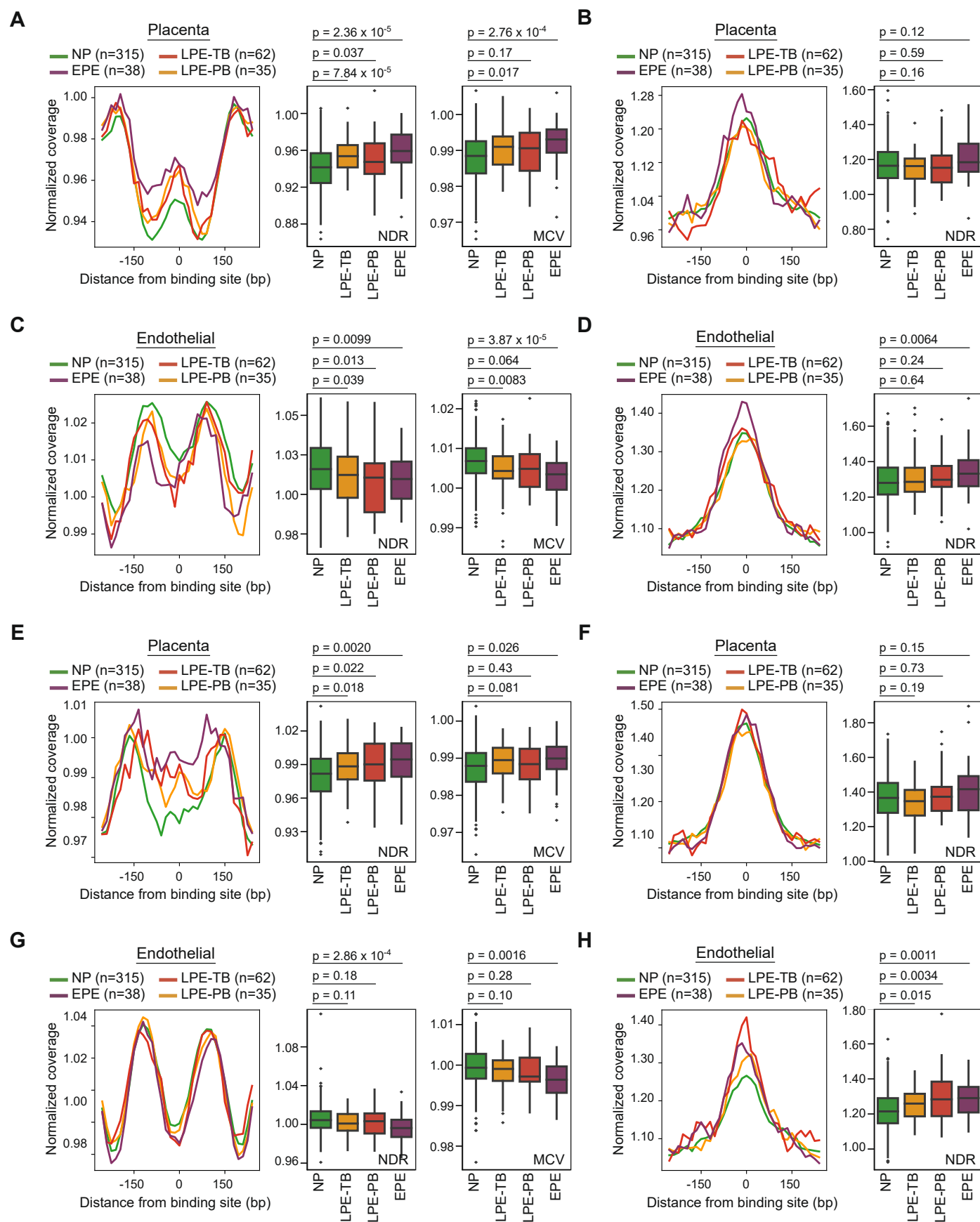
DNase-seq for 20 non-pregnant and 375 PDNAS samples from FF-Training Cohort stratified by GA in weeks of plasma collection. Correlation of tissue values is measured using Pearson's r . The box represents the interquartile range (IQR, 25th and 75th percentiles) with the horizontal line indicating median. The whiskers extend to 1.5X the IQR on either side, with dots representing outliers. *PDNAS*, prenatal cell-free DNA (cfDNA) screening; *GA*, gestational age; *NDR*, nucleosome depleted region.



Extended Data Fig. 6 | See next page for caption.

Extended Data Fig. 6 | Nucleosome profiles for placental- and immune-specific accessible sites by maternal body mass index, derived from scATAC-seq dataset. (a) Nucleosome profile for placenta specific accessible sites derived from scATAC-seq dataset for 354 PDNAS samples (excluding 21 samples with missing BMI) from FF-Training Cohort stratified by BMI at PDNAS. Note decrease in nucleosome coverage (120–180 bp) indicates increased placental tissue contribution. Increase in sub-nucleosome coverage (35–80 bp) indicates increased tissue contribution. (b) Nucleosome profile for immune specific accessible sites derived from scATAC-seq dataset for 354 PDNAS samples (excluding 21 samples with missing BMI) from FF-Training Cohort stratified by BMI at PDNAS. Note decrease in nucleosome coverage (120–180 bp) indicates increased placental tissue contribution. Increase in sub-nucleosome coverage

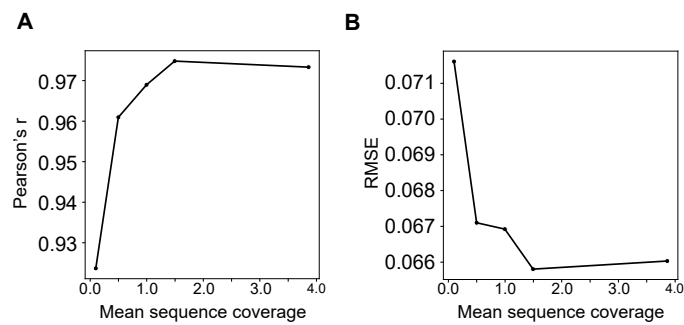
(35–80 bp) indicates increased tissue contribution. Shading represents the interquartile range (IQR, 25th and 75th percentiles) with bold line representing median. (c) Boxplot of nucleosome depleted regions (NDR) of nucleosome (top) and sub-nucleosome (bottom) at tissue specific accessible sites derived from scATAC-seq for 354 PDNAS samples (excluding 21 samples with missing BMI) from FF-Training Cohort stratified by BMI at plasma sampling. Correlation of tissue values is measured using Pearson's r . The box represents the interquartile range (IQR, 25th and 75th percentiles) with the horizontal line indicating median. The whiskers extend to 1.5X the IQR on either side, with dots representing outliers. *PDNAS*, prenatal cell-free DNA (cfDNA) screening; *BMI*, body mass index; *NDR*, nucleosome depleted region.



Extended Data Fig. 7 | See next page for caption.

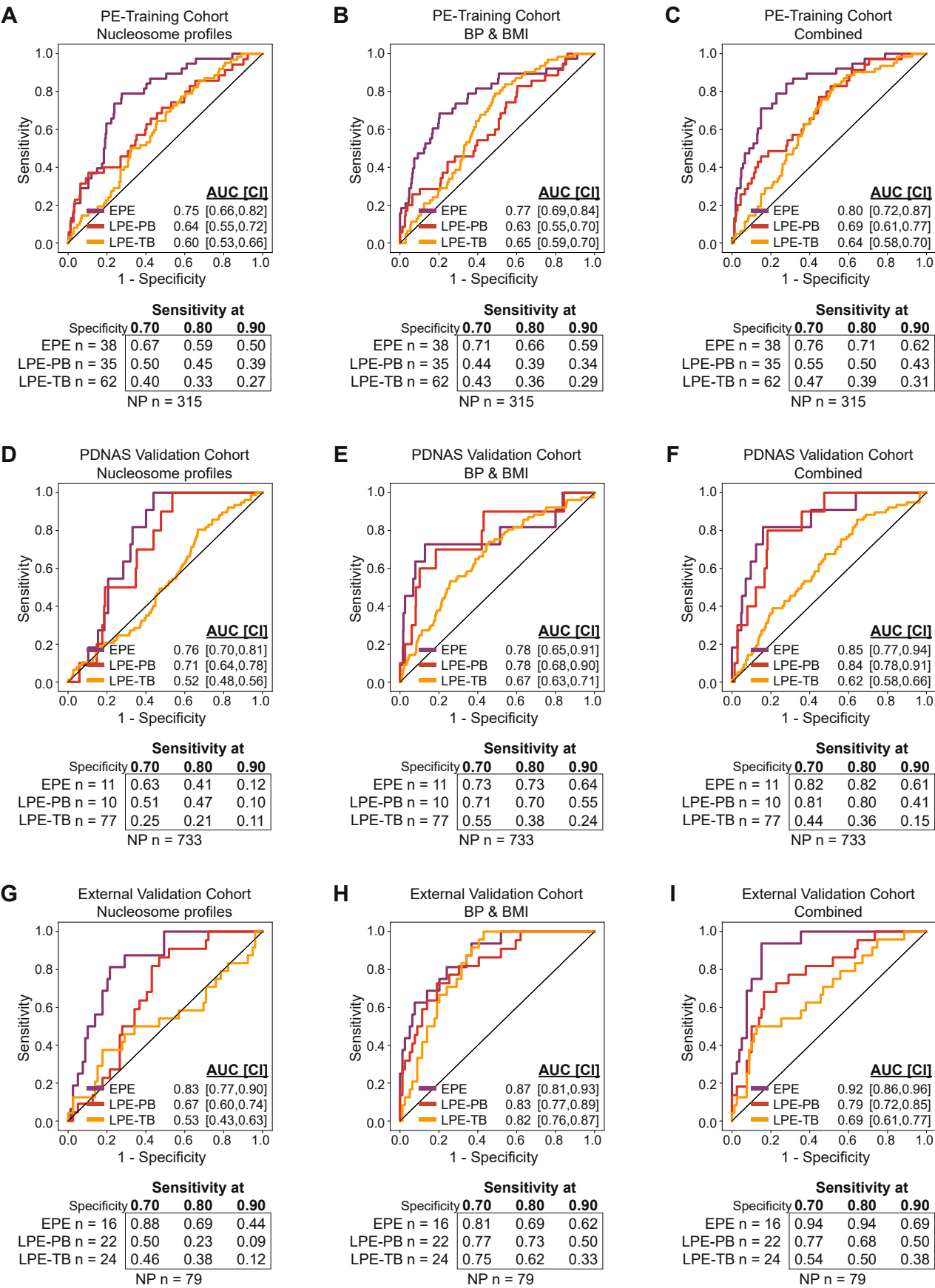
Extended Data Fig. 7 | Placental and endothelial nucleosome and subnucleosome profiles based on preeclampsia phenotype. (a) Placental nucleosome profile (left) and NDR/MCV bar plot (right) from scATAC-seq derived sites for PDNAS samples from the PE-Training Cohort (n = 450). Note higher NDR/MCV values in samples that developed PE indicates decreased placental contribution. (b) Placental sub-nucleosome profile (left) and NDR bar plot (right) from scATAC-seq derived sites for PDNAS samples from the PE-Training Cohort (n = 450). (c) Endothelial nucleosome profile (left) and NDR/MCV bar plot (right) from scATAC-seq derived sites for PDNAS samples from the PE-Training Cohort (n = 450). Note lower NDR/MCV values in samples that developed PE indicates increased endothelial contribution. (d) Endothelial sub-nucleosome profile (left) and NDR bar plot (right) from scATAC-seq derived sites for PDNAS samples from the PE-Training Cohort (n = 450). Note higher NDR values in samples that developed PE indicates increased endothelial contribution. (e) Placental nucleosome profile (left) and NDR/MCV bar plot (right) from DNase-seq derived sites for PDNAS samples from the PE-Training Cohort (n = 450). Note higher

NDR/MCV values in samples that developed PE indicates decreased placental contribution. (f) Placental sub-nucleosome profile (left) and NDR bar plot (right) from DNase-seq derived sites for PDNAS samples from the PE-Training Cohort (n = 450). (g) Endothelial nucleosome profile (left) and NDR/MCV bar plot (right) from DNase-seq derived sites for PDNAS samples from the PE-Training Cohort (n = 450). Note lower NDR/MCV values in samples that developed PE indicates increased endothelial contribution. (h) Endothelial sub-nucleosome profile (left) and NDR bar plot (right) from DNase-seq derived sites for PDNAS samples from the PE-Training Cohort (n = 450). Note higher NDR values in samples that developed PE indicates increased endothelial contribution. The box represents the interquartile range (IQR, 25th and 75th percentiles) with the horizontal line indicating median. The whiskers extend to 1.5X the IQR on either side, with dots representing outliers. Statistical comparison was done using two-sided Mann-Whitney U test. *PE*, preeclampsia; *PDNAS*, prenatal cell-free DNA (cfDNA) screening; *NDR*, nucleosome depleted region; *MCV* mean central coverage.



Extended Data Fig. 8 | Fetal fraction prediction using nucleosome profiling at various sequencing depths. (a) Predicted FF Pearson's r performance at various sequencing depths (Median; 4x, 1.5x, 1x, 0.5x and 0.1x) for 70 FF-External

validation samples. **(b)** Predicted FF RMSE performance at various sequencing depths (Median; 4x, 1.5x, 1x, 0.5x and 0.1x) for 70 FF-External validation samples. *FF*, fetal fraction; *RMSE*, root mean square error.



Extended Data Fig. 9 | See next page for caption.

Extended Data Fig. 9 | Preeclampsia risk prediction among the three cohorts using nucleosome and clinical features alone and in combination. (a) ROC curves for predicting risk of developing PE using nucleosome profiling features only for samples collected at ≤ 16 weeks GA from the PE-Training Cohort ($n = 450$). (b) ROC curves for predicting risk of developing PE using BP and BMI only for samples collected at ≤ 16 weeks GA from the PE-Training Cohort ($n = 450$). (c) ROC curves for predicting risk of developing PE using nucleosome profiling, BP, and BMI for samples collected at ≤ 16 weeks GA from the PE-Training Cohort ($n = 450$). (d) ROC curves for predicting risk of developing PE using nucleosome profiling features only for samples collected at ≤ 16 weeks GA from the PDNAS Validation Cohort ($n = 831$). (e) ROC curves for predicting risk of developing PE using BP and BMI only for samples collected at ≤ 16 weeks GA from the PDNAS Validation Cohort ($n = 831$). (f) ROC curves for predicting risk of developing PE using nucleosome profiling, BP, and BMI for samples collected at ≤ 16 weeks

GA from the PDNAS Validation Cohort ($n = 831$). (g) ROC curves for predicting risk of developing PE using nucleosome profiling features only for samples collected at ≤ 16 weeks GA from the External Validation Cohort ($n = 141$). (h) ROC curves for predicting risk of developing PE using BP and BMI only for samples collected at ≤ 16 weeks GA from the External Validation Cohort ($n = 141$). (i) ROC curves for predicting risk of developing PE using nucleosome profiling, BP, and BMI for samples collected at ≤ 16 weeks GA from the External Validation Cohort ($n = 141$). Performance metrics calculated such as sensitivity and AUC (95% Confidence interval), generated using a bootstrapping with replacement framework with 1000 iterations. *PE*, preeclampsia; *PDNAS*, prenatal cell-free DNA (cfDNA) screening; *GA*, gestational age; *NP*, normal pregnancy; *LPE-TB*, late preeclampsia with term birth; *LPE-PB*, late preeclampsia with preterm birth; *EPE*, early preeclampsia.

Extended Data Table 1 | Demographics of the Training and Validation Cohorts Source data

	PE-Training Cohort (n=450)			PDNAS Validation Cohort (n=831)			External Validation Cohort (n=141)		
	Normal	Preeclampsia	p-value	Normal	Preeclampsia	p-value	Normal	Preeclampsia	p-value
Total	315	135	-	733	98	-	79	62	-
Maternal age at delivery (years) ^a	34.7 ± 3.9	35.5 ± 4.4	0.045	35.2 ± 4.0	35.6 ± 3.9	0.44	-	-	-
BMI at sample collection (kg/m ²) ^a	26.0 ± 6.0	29.1 ± 8.1	2.88 x 10⁻⁵	26.0 ± 5.5	28.0 ± 7.1	0.0016	26.6 ± 6.9	31.4 ± 8.2	3.27 x 10⁻⁵
Maternal race ^b									
White	205 (65.1)	87 (64.4)	0.98	501 (68.3)	62 (63.3)	0.37	58 (73.4)	35 (56.5)	0.053
Asian	73 (23.2)	23 (17.0)	0.18	131 (17.9)	21 (21.4)	0.47	10 (12.7)	11 (17.7)	0.55
Black / African American	14 (4.4)	14 (10.4)	0.03	35 (4.8)	9 (9.2)	0.11	5 (6.3)	13 (21.0)	0.02
American Indian / Alaska Native	4 (1.3)	2 (1.5)	1	18 (2.5)	2 (2.0)	1	-	1 (1.6)	-
Native Hawaiian / Pacific Islander	-	-	-	5 (0.7)	-	-	-	-	-
More than one	1 (0.3)	-	-	7 (1.0)	-	-	4 (5.1)	-	-
Missing	18 (5.7)	9 (6.7)	-	36 (4.9)	4 (4.1)	0.91	1 (1.3)	1 (1.6)	0.9
Hispanic or Latino ^b	24 (7.6)	8 (5.9)	1	92 (12.6)	6 (6.1)	0.092	15 (19.0)	20 (32.3)	0.11
Gestational age at sample collection (weeks) ^b	12.2 ± 1.2	12.2 ± 1.2	0.77	12.2 ± 1.1	12.3 ± 0.9	0.1	12.4 ± 1.1	11.9 ± 1.3	0.048
Gestational age at preeclampsia diagnosis (weeks) ^a	-	35.6 ± 3.7	-	-	37.4 ± 3.2	-	-	35.1 ± 3.6	-
Gestational age at delivery (weeks) ^a	39.4 ± 1.2	36.2 ± 3.2	1.28 x 10⁻³¹	39.2 ± 1.6	37.8 ± 2.8	5.60 x 10⁻⁸	39.8 ± 1.1	35.8 ± 3.1	1.52 x 10⁻¹⁶
Neonatal birth weight (grams) ^a	3448 ± 431	2628 ± 871	1.54 x 10⁻²³	3364 ± 537	3053 ± 805	5.65 x 10⁻⁴	3485 ± 477	2557 ± 58	6.23 x 10⁻¹¹
Male neonatal sex ^b	170 (54.0)	68 (50.4)	0.66	402 (54.8)	60 (61.2)	0.28	40 (50.6)	38 (61.3)	0.27
Female neonatal sex ^b	145 (46.0)	67 (49.6)	-	331 (45.2)	39 (38.8)	-	39 (49.4)	24 (38.7)	-
Chronic Hypertension ^b	19 (6.0)	28 (20.7)	6.57 x 10⁻⁶	34 (4.6)	17 (17.3)	2.61 x 10⁻⁶	1 (1.3)	16 (25.8)	2.90 x 10⁻⁵
Pregestational Diabetes ^b	6 (1.9)	11 (8.1)	0.0036	33 (4.5)	8 (8.2)	0.19	-	5 (8.1)	-
Preeclampsia type									
Early preeclampsia (EPE) ^b	-	38 (8.4)	-	-	11 (13)	-	-	16 (11.3)	-
Late preeclampsia with preterm birth (LPE-PB) ^b	-	35 (7.8)	-	-	10 (1.2)	-	-	22 (15.6)	-
Late preeclampsia with term birth (LPE-TB) ^b	-	62 (13.8)	-	-	77 (9.3)	-	-	24 (17.0)	-

^a, mean ± standard deviation.^b, n (%)

BMI, body mass index; GA, gestational age; LPE-TB, late preeclampsia with term-birth; LPE-PB, late preeclampsia with preterm-birth; EPE, early preeclampsia

bold formatting for statistical significance (P-value < 0.05)

Maternal characteristics and obstetric outcomes for those with preeclampsia or normal pregnancies for each distinct cohort utilized in this study.

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

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☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☐ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Clinical data was abstracted from the medical record or obtained from the respective biorepositories. PDNAS sequence data is housed within our institution. The external samples were sequenced at an external site (The Broad Institute, Boston, MA, USA) and BAM files were provided to our study team.
Data analysis	All sample sequence data was aligned to reference genome hg38 (GRCh38, https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000001405.26/). Raw fastq were first adapter trimmed using cutadapt tool (v3.3) 42. The paired reads were then aligned using BWA-MEM algorithm (v0.7.17) and further processed using SAMtools (v1.10) to sort and index the resulting bam file 43,44. Picard tool's (v2.18.29, http://broadinstitute.github.io/picard) MarkDuplicates function was used to mark duplicates. Picard tool's CollectAlignmentSummaryMetrics, CollectWgsMetrics and CollectInsertSizeMetrics was used to calculate QC sequencing metrics. Custom code, scripts, and models for the PEARL framework can be accessed at https://github.com/GavinHaLab/PEARL/. The Griffin software used in this study can be accessed at https://github.com/GavinHaLab/Griffin/releases/tag/v0.2.0.

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All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The data necessary to reproduce the findings in this study can be accessed at <https://github.com/GavinHaLab/PEARL/> and the Supplementary Tables. Cell-free DNA sequence data is housed at the University of Washington and given the privacy restrictions, reuse of these genomic data may be possible upon reasonable request pending approval by the University of Washington Agreements Office, including execution of a Data Transfer and Use Agreement. Inquiries for data usage can be directed to RS (shreer23@uw.edu) and a response is expected within two weeks. Sequence data was aligned to reference genome hg38 (GRCh38, https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000001405.26/). Tissue specific binding sites were derived from the following accessibility datasets: <https://www.meuleman.org/research/dhsindex/>, <http://catlas.org/humanenhancer>.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	Only individuals of female biologic sex were included in this study given the specific conditions of interest (pregnancy, preeclampsia). The proportion of pregnancies carrying a male fetus are presented in Extended Data Table 1.
Reporting on race, ethnicity, or other socially relevant groupings	Participant race and ethnicity were abstracted from the medical record and are based on self-reporting. For the External Validation Cohort, race and ethnicity were provided by the respective biorepositories. No other socially relevant groupings were considered for our analyses.
Population characteristics	Data derives from the population of pregnant individuals carrying a singleton gestation seeking prenatal cell-free DNA screening (PDNAS) for aneuploidy through the University of Washington (UW) between 2017 and 2021. The UW is a quaternary referral center with a catchment area that spans a five state region (Washington, Wyoming, Alaska, Montana, and Idaho). The samples used in the FF-Training Cohort (n=395), the clinical outcomes were purposely blinded to avoid model overfitting. Only samples collected at or prior to 16 weeks gestation were use for training and validating the preeclampsia prediction model. We included samples collected after 16 weeks gestation for fetal fraction tissue of origin analyses. Relevant population characteristics are included in Table 1, including age, body mass index, self-reported race and ethnicity, and rates of chronic hypertension, and preexisting diabetes in our population.
Recruitment	Data for the Training and PDNAS Validation sequencing data was obtained from clinically obtained PDNAS samples. Participants were not specifically recruited for this study. We included all eligible samples meeting inclusion criteria during the specific dates. As such, we do not anticipate any biases to be present. For the External Validation Samples (n=146), informed consent was obtained by the respective biorepositories as required.
Ethics oversight	Ethics oversight was by the University of Washington (STUDY0001636, STUDY00005540, STUDY00014943, STUDY00004878, STUDY00017462).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Life sciences study design

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Sample size	No statistical method was used to predetermine sample size.
Data exclusions	For those with confirmed clinical outcomes, before use of or processing of any PDNAS samples, we excluded those with miscarriage, pregnancy termination, stillbirth, or if the clinical outcome could not be determined based on the available medical records. However, we did not exclude cases of termination or stillbirth if these cases were complicated by preeclampsia (these cases were included in the preeclampsia cases).

For the participants designated as having a normal pregnancy outcome, we excluded those whose pregnancy ended before 24 weeks, if it was complicated by preeclampsia (including postpartum preeclampsia), gestational hypertension, maternal transplant, maternal malignancy, or if there was fetal aneuploidy or a major anomaly (neurologic, cardiac, pulmonary, genitourinary, skeletal, etc.). Notably, we did not exclude those with chronic hypertension, pregestational diabetes, other maternal medical conditions (e.g., autoimmune disease), or small for gestational age infant.

Preeclampsia cases were designated as such based on current clinical guidelines (American College of Obstetricians & Gynecologists Practice Bulletin 222: Gestational Hypertension and Preeclampsia). We excluded cases with postpartum preeclampsia, gestational hypertension, maternal transplant, maternal malignancy, or if the pregnancy was complicated by aneuploidy or major fetal anomaly (see above). For patients with chronic hypertension, we only included cases with superimposed preeclampsia (i.e., those with chronic hypertension exacerbation without preeclampsia were included in the normal cohort).

Replication	The Fetal Fraction (FF) model was validated in an internal cohort (PDNAS Validation Cohort, n=956) and an external cohort (n=70) The preeclampsia prediction model (PEARL) was validated in an internal cohort (PDNAS Validation Cohort, n=956) and the External Validation Cohort (n=146).
Randomization	The study participants were not randomized.
Blinding	All samples were de-identified prior to sharing with the analytic team.

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☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A